

NAGALAND LITERACY AND NUMERACY FEST 2026

Micro-Improvement Project (Science Teacher) Grade 7

Chapter 7 — Heat Transfer

Why Micro-Improvement Project (MIP)?

Across Nagaland, students already live with hotness and coldness every day: the warmth of a hearth in a Morung on a cold Kohima morning, the tin roof that turns a room into an oven by midday, the cool water carried in a bamboo pipe, the thermos flask that keeps tea hot through a long walk to school, and the shawl pulled tight against the December wind. This Micro-Improvement Project (MIP) helps students move from simply saying something 'feels hot' or 'feels cold' toward measuring temperature reliably, reading real thermometers with care, and explaining why some materials and some designs keep heat in or out — using the language of conduction, convection and radiation. It builds directly on the measurement foundations from earlier MIP cycles, and prepares students to reason like young scientists about the everyday hot and cold of their own homes and villages.

Skill in Focus	Heat Transfer — Temperature, Thermometers, Conductors, Insulators and Modes of Heat Transfer
Duration	1 Week
Competency Addressed	C-7.1: “Conducts simple investigations and uses evidence, measurement and everyday observation to explain natural phenomena.”

Learning Outcome

Concept of Temperature and Modes of Heat Transfer

1. Distinguish hotness and coldness by a reliable measure of temperature rather than by touch alone.
2. Distinguish the clinical thermometer from the laboratory thermometer by range, least count and units of measurement.
3. Examine the need for a laboratory thermometer while carrying out experiments in the laboratory.

4. List precautions to be followed while using a clinical and a laboratory thermometer, and identify the role of the kink.
5. Observe the heating and cooling of objects in order to describe conduction.
6. Devise an activity, or collaborate on a situation, to show the rate of thermal conduction, convection and radiation.
7. Apply the concept of convection to the heating of land and water in order to predict and describe land breeze and sea breeze.
8. Explain why a substance remains at the same temperature inside a thermos flask or vacuum bottle.
9. Correlate the modes of transfer of heat with the use of different kinds of clothing in different parts of the world (polar, temperate, and Nagaland's own seasons).

Key Concepts

- Demonstrates that temperature, not touch, is the reliable measure of hotness and coldness.
- Explains the range, least count, unit and correct use of clinical and laboratory thermometers, including the role of the kink.
- Describes conduction, convection and radiation as three distinct ways heat travels, using everyday conductors and insulators.
- Applies convection to explain land breeze and sea breeze, and radiation to explain colour-based heat absorption.
- Explains how a thermos flask blocks all three modes of heat transfer to keep its contents at a steady temperature.
- Correlates modes of heat transfer with clothing and housing choices across different climates, including Nagaland's own seasons.

Detailed Tasks

Task-1

Hot or Cold? Trusting a Real Measure

Objective

Build awareness that touch is an unreliable judge of hotness and coldness, and introduce temperature as a reliable, numerical measure.

Setting	Indoor (Classroom) — a veranda may be used for the water containers
Teacher's Note	Have the three containers of water ready and at noticeably different temperatures before class begins, so the activity starts immediately.
Duration	30 minutes
Materials Needed	Three mugs or small buckets (one cold water, one room-temperature water, one lukewarm water) · A thermometer (clinical, laboratory, or any available in school) · Notebook and pencil

Activity Steps

1. Place the three containers of cold, room-temperature and lukewarm water in front of the class. Ask one student to dip the left hand in the cold water and the right hand in the lukewarm water for ten seconds, then dip both hands into the room-temperature water and describe what each hand feels.
2. Discuss as a class: the same room-temperature water felt cool to one hand and warm to the other. Ask: 'Can we trust our hands alone to say how hot or cold something is?'
3. Introduce the thermometer as an instrument that gives one reliable number regardless of whose hand touches the object. Measure and record the actual temperature of each container.
4. In pairs, students measure the temperature of two or three more everyday objects in the room (a windowsill, a metal almirah, a wooden desk, drinking water) and record the readings in a two-column table: Object, Temperature.
5. Close with a class discussion: 'Why do we need a number, and not just a feeling, to describe how hot or cold something is?'

Concept Built: *Temperature is a reliable, numerical measure of hotness and coldness, more dependable than touch.*

Task-2

Meet the Thermometers — Clinical and Laboratory

Objective	Distinguish the clinical thermometer from the laboratory thermometer by range, least count and units; examine why a laboratory needs a different thermometer; list precautions and identify the role of the kink.
Setting	Indoor (Classroom)
Teacher's Note	If only one type of thermometer is available, use a clearly labelled diagram on the board for the other, so every group can still compare both types side by side.
Duration	40 minutes
Materials Needed	A clinical thermometer (digital, or mercury/alcohol if available and safe) · A laboratory thermometer (0–100°C range) if available, or a large labelled diagram · Chart paper · Magnifying glass (optional)

Activity Steps

1. Show students a clinical thermometer and, if available, a laboratory thermometer side by side. If only one is available, use a clearly labelled diagram on the board for the other.
2. In pairs, students examine (or study the diagram of) each thermometer and record in a table: Range, Smallest division (least count), Unit of measurement, Where it is normally used.
3. Guide students to notice that the clinical thermometer has a narrow range around normal body temperature (about 35°C to 42°C) with a kink just above the bulb, while the laboratory thermometer has a much wider range (often –10°C to 110°C) and no kink.
4. Ask: 'Why can we not use a laboratory thermometer to check a patient's fever, and why can we not use a clinical thermometer to check the temperature of boiling water in an experiment?' Discuss reasons — mismatched range, risk of breakage, hygiene, and the absence of a kink to hold the reading.
5. Introduce the kink: a narrow bend in the tube just above the mercury bulb of a clinical thermometer that stops the liquid from flowing back immediately, so the reading can still be read after the thermometer is removed from the mouth or armpit.
6. In groups, students list at least four precautions for the safe and accurate use of a thermometer — for example, do not bite a mercury thermometer, shake the reading down before use, hold a laboratory thermometer at eye level to read it, avoid touching the bulb while reading, and clean the thermometer before and after use.

Concept Built: *Clinical and laboratory thermometers differ in range, least count and design (the kink); each is suited to a different purpose, and careful handling is essential for safety and accuracy.*

Task-3

Feel the Heat — Conduction, Conductors and Insulators

Objective	Observe the heating and cooling of objects to describe conduction, and classify materials as good or bad conductors of heat.
Setting	Indoor, with an optional Outdoor step if sunlight is available
Teacher's Note	Pour the hot water for each group yourself, and remind students to touch only the dry, upper end of each object — never the end sitting in the hot water.
Duration	35 minutes
Materials Needed	A metal spoon, a wooden stick or spoon, and a plastic spoon or ruler (one set per group) · A cup of hot water, poured by the teacher · Notebook

Activity Steps

1. Ask: 'Why are cooking pots made of metal, but their handles made of wood or plastic?' Take a few guesses from the class before starting the activity.
2. Give each group a metal spoon, a wooden stick, and a plastic spoon. Place all three in a cup of hot water at the same time, with the handles resting outside the cup.
3. After two to three minutes, ask one student in each group to carefully touch the upper, dry end of each object and rank them from hottest to least hot. Record the observations.
4. Explain conduction: heat travels through the particles of a solid from the hotter end to the cooler end without the material itself moving; metals conduct heat quickly (good conductors), while wood, plastic and cloth conduct heat slowly (bad conductors, or insulators).
5. Optional outdoor step: on a sunny day, place a metal object and a wooden object of similar size on the school ground for ten minutes and compare how hot each becomes to the touch.

6. Discuss: 'Why are tin roofs so hot to the touch by midday, while a thatched or wooden roof feels cooler? Why do we use a cloth or wooden holder to lift a hot cooking pot?'

Concept Built: *Conduction is the transfer of heat through a solid from a hotter point to a cooler one; good conductors (metals) transfer heat quickly, while insulators (wood, plastic, cloth) transfer it slowly.*

Task-4

Rising Smoke, Sinking Cool — Convection and Land-Sea Breeze

Objective	Devise and observe an activity that shows the rate and direction of heat movement in convection, and apply the idea to explain land breeze and sea breeze.
Setting	Indoor — teacher-led demonstration only (open flame and smoke)
Teacher's Note	The teacher alone lights the candle and incense stick; matches and open flame are never handed to students. Use the water-and-food-colour alternative instead if an open flame is not permitted in your classroom.
Duration	35 minutes
Materials Needed	A candle · An incense stick · A matchbox · (Flame-free alternative) a tall glass of hot water, a tall glass of cold water, and a drop of food colour

Activity Steps

1. Caution: the teacher alone lights the candle and incense stick; matches and open flame are never handed to learners.
2. Light the candle and hold a lit incense stick a little above and to the side of the flame. Ask students to watch the direction the smoke moves.
3. Students observe that the smoke is pulled toward the candle and then rises upward. Discuss: air near the candle becomes hot, becomes lighter, and rises; cooler air moves in from the sides to take its place — this continuous movement of heat through a gas (or a liquid) is called convection.
4. If an open flame cannot be used, demonstrate with water instead: place one drop of food colour gently into a glass of hot water and a separate glass of cold water, and compare how quickly the colour spreads in each (it spreads faster in hot water because of the same rising-and-mixing movement).

5. Draw a simple diagram on the board showing daytime and night-time air movement near a river, pond, or lake close to the school or village: during the day, land heats up faster than water, so hot air over the land rises and cooler air moves in from above the water — this is why a breeze is often felt near a river or pond by afternoon; at night, land cools faster than water, so the direction reverses.

6. Ask students to predict: 'If a village is next to a large river, would you expect a cooler breeze from the river in the afternoon or at night? Why?'

Concept Built: *Convection is the movement of heat through a rising current of hot gas or liquid and a sinking current of cooler gas or liquid; this same principle explains land breeze and sea (or river) breeze.*

Task-5

Catching the Sun — Radiation and the Colour of Clothes

Objective	Observe and compare the rate of heat transfer by radiation using different colours, and connect this to everyday clothing choices.
Setting	Outdoor — needs a sunny day; reschedule the activity if the sky is overcast
Teacher's Note	Prepare the two wrapped bottles the same morning so both start at the same temperature before going out into the sun.
Duration	30 minutes
Materials Needed	Two identical plastic bottles filled with equal amounts of water · Black paper and white paper to wrap around each bottle · Rubber bands or tape · A thermometer if available, or reliance on careful touch

Activity Steps

1. Ask: 'On a sunny day, why do we feel more comfortable in a white or light-coloured shirt than in a black one?'
2. Wrap one bottle completely in black paper and the other in white paper. Fill both with the same amount of water and place them in direct sunlight for fifteen to twenty minutes.
3. Measure, or carefully compare by touch, the temperature of the water in each bottle. Record which one became hotter.

4. Explain radiation: heat from the sun travels directly through empty space as rays, without needing any material to carry it — unlike conduction and convection, which both need a material medium. Dark surfaces absorb more radiant heat, while light surfaces reflect more of it.

5. Connect to daily life: discuss why Naga elders often wear dark, tightly woven shawls in the cold months (to absorb and hold the sun's warmth), yet people prefer lighter clothes and roofs in hot weather; why some house roofs and walls are painted a lighter colour; why a dark tin roof feels hotter to the touch than a whitewashed wall.

Concept Built: *Radiation is the transfer of heat directly through space as rays, needing no medium; dark surfaces absorb more radiant heat than light surfaces.*

Task-6

The Flask That Keeps Its Cool — Why a Thermos Stays the Same Temperature

Objective	Explain why a substance remains at nearly the same temperature inside a thermos flask (vacuum bottle) by combining the ideas of conduction, convection and radiation learnt so far.
Setting	Indoor (Classroom)
Teacher's Note	Pour both cups of hot water at exactly the same time so the comparison is fair; if a real thermos is not available, use a well-labelled cut-away diagram instead.
Duration	30 minutes
Materials Needed	A real thermos flask or vacuum bottle, or a labelled cut-away diagram on chart paper · Hot water · An ordinary cup and a thermos, for comparison · A clock or timer

Activity Steps

1. Pour equal amounts of hot water into an ordinary cup and into a thermos flask at the same time. Leave both undisturbed.
2. After twenty to thirty minutes (or checked again later in the day), compare which one has cooled down more.
3. Show a labelled diagram of a thermos flask on the board or chart paper: a double-walled glass or steel body with a vacuum (empty space) between the walls, a silvered, shiny inner surface, and a tight-fitting stopper or cap.

4. Guide students to connect each feature to a mode of heat transfer already studied: the vacuum has no material to carry heat, so conduction and convection cannot occur across it; the shiny, silvered surface reflects radiant heat back inward instead of letting it escape; the stopper or cap prevents heat loss through the mouth of the flask by convection of air.

5. Ask: 'Why do families in Nagaland carry tea or water in a thermos on a long walk, at a paddy field, or during the Hornbill Festival, instead of in an ordinary bottle?'

Concept Built: *A thermos flask keeps its contents at nearly the same temperature by blocking all three modes of heat transfer — conduction and convection across a vacuum, and radiation through a silvered, reflective inner surface.*

Task-7

Dressing for the Weather — Heat Transfer and Our Clothes

Objective	Correlate the modes of heat transfer with the kind of clothing worn in different climates of the world, connecting classroom learning to daily life and Naga tradition.
Setting	Homework — done at home with family, discussed in class the next day
Teacher's Note	Explain this clearly one day in advance, so students remember to look at their own clothing and ask a family elder the discussion question.
Duration	20 minutes
Materials Needed	Notebook and pencil · Optional: a family shawl, jacket or winter wrap to examine closely

Activity Steps

1. Ask each student to look at two or three items of clothing worn at home in different seasons — for example, a thick woollen shawl worn in December and a thin cotton shirt worn in the hot months.
2. Students note the colour, thickness and material of each piece of clothing.
3. Using what was learnt about conduction, convection and radiation, students write one or two sentences for each item explaining why it suits that season (for example: thick wool traps air and is a poor conductor, so it keeps body heat in during winter; a dark-coloured shawl helps absorb the sun's warmth; light cotton lets body heat escape easily in summer).

4. Ask an elder at home: 'What do people wear in very cold places like the Himalaya or the polar regions, and what do people wear in very hot places?' Students record the answer and think about which mode of heat transfer explains each choice.
5. The next day, students share their findings with a partner or with the class.

Concept Built: *Transfers the modes of heat transfer to real clothing choices across different climates, strengthening the link between classroom science and daily life.*

Culminating Showcase

Showcase	Nagaland Heat Detectives — Final Project
----------	--

Objective	Bring together every concept from Tasks 1–7 into one creative, real-world design or investigation project that groups plan, build or test, and present.
Setting	Combined Classroom and Outdoor project — 2–3 sessions after Task-7, culminating in a class exhibition
Teacher's Note	Encourage each group to pick a genuine question from their own home or village rather than a generic one — real context makes the investigation and presentation stronger.
Duration	30–35 minutes per session (2–3 sessions)
Materials Needed	Chart paper or cardboard · Plastic bottles, cloth, black and white paper, small stones or sand · Thermometer, if available · Locally available materials such as bamboo, clay or dried leaves — optional

Activity Steps

1. In groups of 4–5, students choose one real problem or question from their own home or village related to heat and temperature — for example, keeping a house cool without electricity, keeping tea or water warm for longer, choosing the right roof colour, or explaining a family cooling or heating tradition.
2. Groups plan a simple model, poster, or investigation that uses at least two of the ideas studied: temperature measurement, thermometers, conduction, convection, radiation, or vacuum insulation.

3. Groups build or sketch their design, or set up their investigation, clearly labelling which mode(s) of heat transfer it uses and why.
4. Groups test their idea where possible (for example, measuring temperature before and after, or comparing two versions) and record their observations on chart paper.
5. In the final session, each group presents its project in a short class exhibition, explaining the science behind their design and one interesting or surprising discovery.

Concept Built: *Independent application of temperature measurement, thermometers, conduction, convection, radiation and insulation in one integrated, locally-grounded investigation or design.*

BY THE END OF THIS PROJECT, TEACHERS ALONG WITH STUDENTS ARE REQUIRED TO COME UP WITH INNOVATIVE PROJECTS THAT CAN BE EXHIBITED WITHIN THE CLASSROOM.

By the end of these tasks, teachers and students should have moved from guessing 'hot' or 'cold' by touch to confidently measuring, reading, and explaining temperature and heat transfer — culminating in an original, locally-grounded investigation or model that each group can showcase with pride.

Reflect on Students' Understanding

Observe 5–6 students each week during the tasks above. No written test is needed — watch how students measure temperature, read a thermometer, and explain their thinking.

Student Name	Reads Temperature Reliably	Distinguishes Thermometers	Explains Conduction, Convection Radiation	Applies Heat Transfer to Daily & Life	Needs Support
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐

Scale (for each column): ☐ Not Yet ☐ Sometimes ☐ Consistently

Picture Guide

For teachers and school artists to adapt into a single classroom poster (in the style of the Chapter 6 Perimeter and Area picture guide), showing all six panels below in one sheet titled 'Heat Transfer in Our Life!'

HEAT TRANSFER IN OUR LIFE!

1. HOT OR COLD?

Touch fools us — the same water feels warm to one hand, cool to the other.

A thermometer gives one true number.

2. MEET THE THERMOMETERS

Clinical (35°C–42°C, has a kink) vs Laboratory (–10°C to 110°C, no kink).

Different range, different job.

3. FEEL THE HEAT

A metal spoon in hot water heats up fast (good conductor).

Wood and plastic stay cool longer (insulators).

4. RISING SMOKE

Smoke bends toward the candle, then rises.

Hot air rises, cool air sinks — this is convection.

5. CATCHING THE SUN

The black-wrapped bottle gets hotter than the white one in sunlight.

Dark colours absorb more radiation.

6. THE FLASK THAT KEEPS ITS COOL

Vacuum stops conduction & convection; the silvered wall reflects radiation back in.

Tea stays hot for hours.

Picture Guide

HEAT TRANSFER IN OUR LIFE

MEASURE • OBSERVE • UNDERSTAND

1. TEMPERATURE



Temperature is a measure of how hot or cold something is.

2. THERMOMETERS

Clinical Thermometer
(for body)



35°C - 42°C
°C

Laboratory Thermometer
(for experiments)



0°C - 100°C
°C

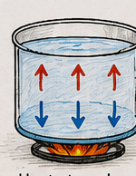
3. HEAT TRANSFER

Conduction
(By touch)



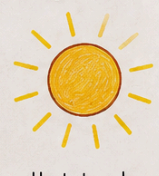
Heat travels through solids.

Convection
(By movement)



Heat travels through liquids and gases.

Radiation
(By waves)



Heat travels through empty space.

4. EVERYDAY EXAMPLES



Warmth from hearth (conduction and radiation)



Tin roof gets hot by radiation



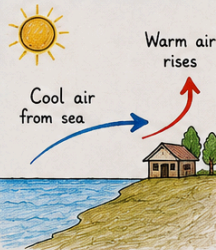
Thermos flask keeps hot by stopping heat transfer



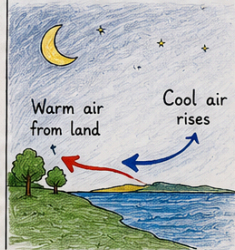
Woollen clothes keep us warm (less heat loss)

5. LAND & SEA BREEZE

DAY (SEA BREEZE)



NIGHT (LAND BREEZE)



6. GOOD INSULATORS

Wood



Wool



Air



Do not allow heat to pass easily.



HEAT TRAVELS IN THREE WAYS - CONDUCTION, CONVECTION AND RADIATION.

UNDERSTAND HEAT, USE HEAT, SAVE ENERGY!